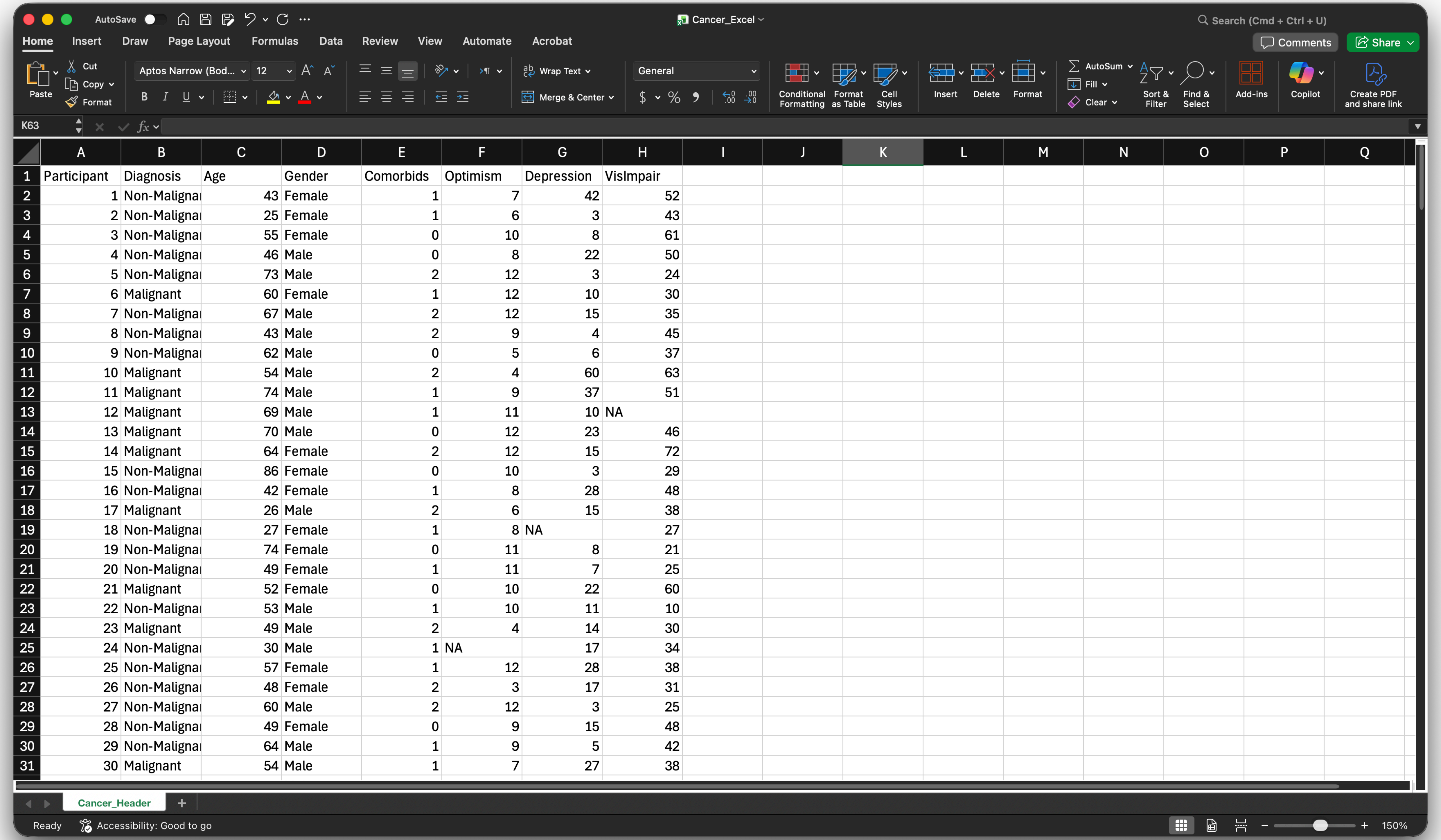
LAB WEEK 10

IMPORTING DATA INTO R

DATA FILE FEATURES

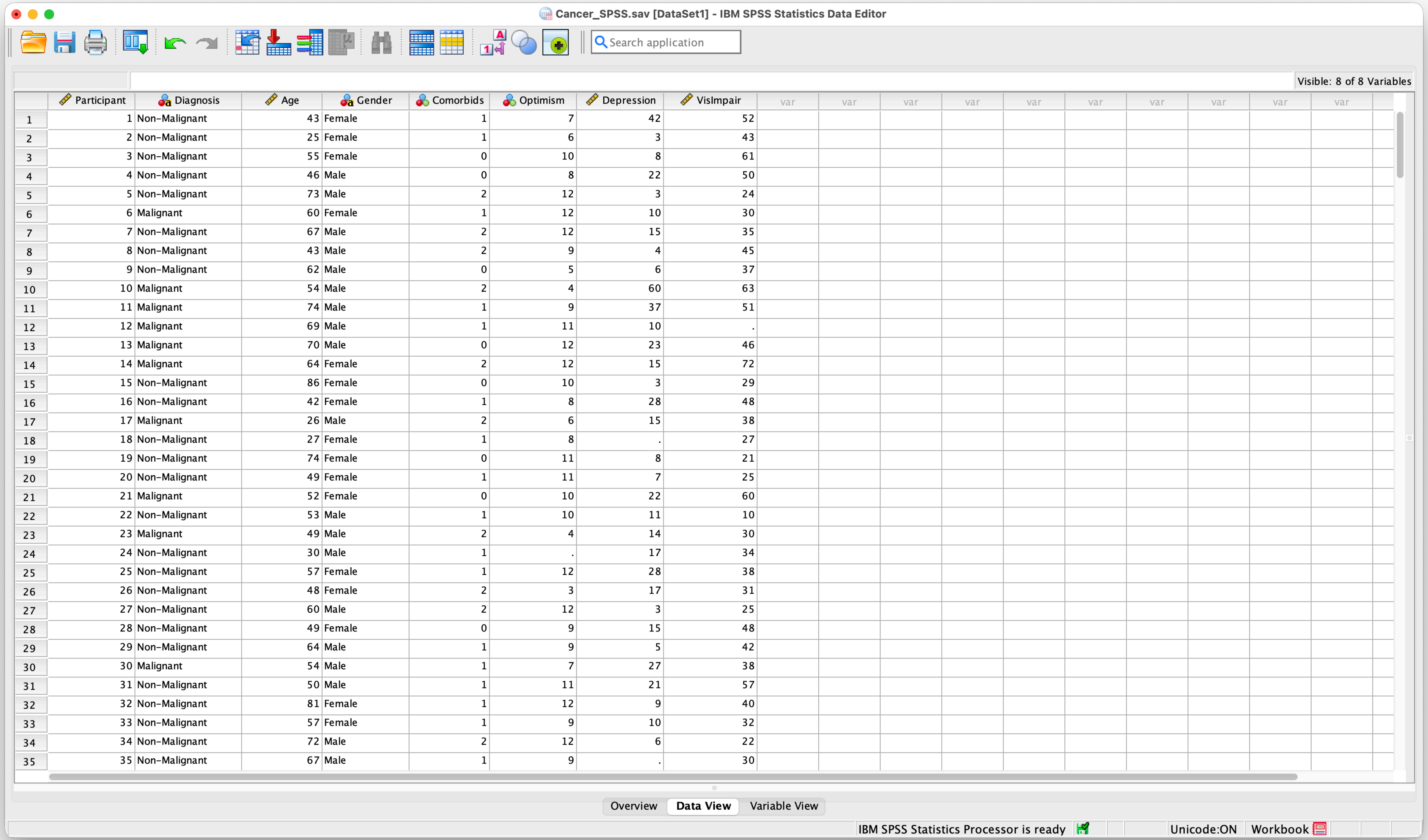
- Proprietary format (Excel) or universal plain text format (.csv, .dat, or .txt)?
- Are variable names in the first row (header)?
- In universal plain text format, are variables in columns separated by commas or spaces?

PROPRIETARY FORMAT: EXCEL



Participant	Diagnosis	Age	Gender	Comorbids	Optimism	Depression	VisImpair
1	Non-Maligna	43	Female	1	7	42	52
2	Non-Maligna	25	Female	1	6	3	43
3	Non-Maligna	55	Female	0	10	8	61
4	Non-Maligna	46	Male	0	8	22	50
5	Non-Maligna	73	Male	2	12	3	24
6	Malignant	60	Female	1	12	10	30
7	Non-Maligna	67	Male	2	12	15	35
8	Non-Maligna	43	Male	2	9	4	45
9	Non-Maligna	62	Male	0	5	6	37
10	Malignant	54	Male	2	4	60	63
11	Malignant	74	Male	1	9	37	51
12	Malignant	69	Male	1	11	10	NA
13	Malignant	70	Male	0	12	23	46
14	Malignant	64	Female	2	12	15	72
15	Non-Maligna	86	Female	0	10	3	29
16	Non-Maligna	42	Female	1	8	28	48
17	Malignant	26	Male	2	6	15	38
18	Non-Maligna	27	Female	1	8	NA	27
19	Non-Maligna	74	Female	0	11	8	21
20	Non-Maligna	49	Female	1	11	7	25
21	Malignant	52	Female	0	10	22	60
22	Non-Maligna	53	Male	1	10	11	10
23	Malignant	49	Male	2	4	14	30
24	Non-Maligna	30	Male	1	NA	17	34
25	Non-Maligna	57	Female	1	12	28	38
26	Non-Maligna	48	Female	2	3	17	31
27	Non-Maligna	60	Male	2	12	3	25
28	Non-Maligna	49	Female	0	9	15	48
29	Non-Maligna	64	Male	1	9	5	42
30	Malignant	54	Male	1	7	27	38

PROPRIETARY FORMAT: SPSS



	Participant	Diagnosis	Age	Gender	Comorbids	Optimism	Depression	VisImpair
1	1	Non-Malignant	43	Female	1	7	42	52
2	2	Non-Malignant	25	Female	1	6	3	43
3	3	Non-Malignant	55	Female	0	10	8	61
4	4	Non-Malignant	46	Male	0	8	22	50
5	5	Non-Malignant	73	Male	2	12	3	24
6	6	Malignant	60	Female	1	12	10	30
7	7	Non-Malignant	67	Male	2	12	15	35
8	8	Non-Malignant	43	Male	2	9	4	45
9	9	Non-Malignant	62	Male	0	5	6	37
10	10	Malignant	54	Male	2	4	60	63
11	11	Malignant	74	Male	1	9	37	51
12	12	Malignant	69	Male	1	11	10	.
13	13	Malignant	70	Male	0	12	23	46
14	14	Malignant	64	Female	2	12	15	72
15	15	Non-Malignant	86	Female	0	10	3	29
16	16	Non-Malignant	42	Female	1	8	28	48
17	17	Malignant	26	Male	2	6	15	38
18	18	Non-Malignant	27	Female	1	8	.	27
19	19	Non-Malignant	74	Female	0	11	8	21
20	20	Non-Malignant	49	Female	1	11	7	25
21	21	Malignant	52	Female	0	10	22	60
22	22	Non-Malignant	53	Male	1	10	11	10
23	23	Malignant	49	Male	2	4	14	30
24	24	Non-Malignant	30	Male	1	.	17	34
25	25	Non-Malignant	57	Female	1	12	28	38
26	26	Non-Malignant	48	Female	2	3	17	31
27	27	Non-Malignant	60	Male	2	12	3	25
28	28	Non-Malignant	49	Female	0	9	15	48
29	29	Non-Malignant	64	Male	1	9	5	42
30	30	Malignant	54	Male	1	7	27	38
31	31	Non-Malignant	50	Male	1	11	21	57
32	32	Non-Malignant	81	Female	1	12	9	40
33	33	Non-Malignant	57	Female	1	9	10	32
34	34	Non-Malignant	72	Male	2	12	6	22
35	35	Non-Malignant	67	Male	1	9	.	30

COMMA DELIMITED WITH HEADER

Cancer_Header.csv

```
Participant,Diagnosis,Age,Gender,Comorbids,Optimism,Depression,VisImpair
1,Non-Malignant,43,Female,1,7,42,52
2,Non-Malignant,25,Female,1,6,3,43
3,Non-Malignant,55,Female,0,10,8,61
4,Non-Malignant,46,Male,0,8,22,50
5,Non-Malignant,73,Male,2,12,3,24
6,Malignant,60,Female,1,12,10,30
7,Non-Malignant,67,Male,2,12,15,35
8,Non-Malignant,43,Male,2,9,4,45
9,Non-Malignant,62,Male,0,5,6,37
10,Malignant,54,Male,2,4,60,63
11,Malignant,74,Male,1,9,37,51
12,Malignant,69,Male,1,11,10,NA
13,Malignant,70,Male,0,12,23,46
14,Malignant,64,Female,2,12,15,72
15,Non-Malignant,86,Female,0,10,3,29
16,Non-Malignant,42,Female,1,8,28,48
17,Malignant,26,Male,2,6,15,38
18,Non-Malignant,27,Female,1,8,NA,27
19,Non-Malignant,74,Female,0,11,8,21
20,Non-Malignant,49,Female,1,11,7,25
21,Malignant,52,Female,0,10,22,60
22,Non-Malignant,53,Male,1,10,11,10
23,Malignant,49,Male,2,4,14,30
24,Non-Malignant,30,Male,1,NA,17,34
25,Non-Malignant,57,Female,1,12,28,38
26,Non-Malignant,48,Female,2,3,17,31
27,Non-Malignant,60,Male,2,12,3,25
28,Non-Malignant,49,Female,0,9,15,48
29,Non-Malignant,64,Male,1,9,5,42
30,Malignant,54,Male,1,7,27,38
31,Non-Malignant,50,Male,1,11,21,57
32,Non-Malignant,81,Female,1,12,9,40
33,Non-Malignant,57,Female,1,9,10,32
34,Non-Malignant,72,Male,2,12,6,22
35,Non-Malignant,67,Male,1,9,NA,30
36,Non-Malignant,32,Male,1,3,13,14
37,Non-Malignant,81,Male,1,12,7,10
38,Non-Malignant,83,Female,0,8,13,68
39,Malignant,50,Female,1,12,9,26
40,Malignant,62,Male,2,9,16,65
41,Malignant,68,Female,0,11,25,64
```

Line 46, Column 1 | Tab Size: 4 | Plain Text

COMMA DELIMITED WITHOUT HEADER

Cancer_NoHeader.csv

```
1,Non-Malignant,43,Female,1,7,42,52
2,Non-Malignant,25,Female,1,6,3,43
3,Non-Malignant,55,Female,0,10,8,61
4,Non-Malignant,46,Male,0,8,22,50
5,Non-Malignant,73,Male,2,12,3,24
6,Malignant,60,Female,1,12,10,30
7,Non-Malignant,67,Male,2,12,15,35
8,Non-Malignant,43,Male,2,9,4,45
9,Non-Malignant,62,Male,0,5,6,37
10,Malignant,54,Male,2,4,60,63
11,Malignant,74,Male,1,9,37,51
12,Malignant,69,Male,1,11,10,NA
13,Malignant,70,Male,0,12,23,46
14,Malignant,64,Female,2,12,15,72
15,Non-Malignant,86,Female,0,10,3,29
16,Non-Malignant,42,Female,1,8,28,48
17,Malignant,26,Male,2,6,15,38
18,Non-Malignant,27,Female,1,8,NA,27
19,Non-Malignant,74,Female,0,11,8,21
20,Non-Malignant,49,Female,1,11,7,25
21,Malignant,52,Female,0,10,22,60
22,Non-Malignant,53,Male,1,10,11,10
23,Malignant,49,Male,2,4,14,30
24,Non-Malignant,30,Male,1,NA,17,34
25,Non-Malignant,57,Female,1,12,28,38
26,Non-Malignant,48,Female,2,3,17,31
27,Non-Malignant,60,Male,2,12,3,25
28,Non-Malignant,49,Female,0,9,15,48
29,Non-Malignant,64,Male,1,9,5,42
30,Malignant,54,Male,1,7,27,38
31,Non-Malignant,50,Male,1,11,21,57
32,Non-Malignant,81,Female,1,12,9,40
33,Non-Malignant,57,Female,1,9,10,32
34,Non-Malignant,72,Male,2,12,6,22
35,Non-Malignant,67,Male,1,9,NA,30
36,Non-Malignant,32,Male,1,3,13,14
37,Non-Malignant,81,Male,1,12,7,10
38,Non-Malignant,83,Female,0,8,13,68
39,Malignant,50,Female,1,12,9,26
40,Malignant,62,Male,2,9,16,65
41,Malignant,68,Female,0,11,25,64
42,Non-Malignant,68,Female,2,12,11,32
```

Line 46, Column 1 Tab Size: 4 Plain Text

SPACE DELIMITED WITH HEADER

Cancer_NoHeader.csv

```
1,Non-Malignant,43,Female,1,7,42,52
2,Non-Malignant,25,Female,1,6,3,43
3,Non-Malignant,55,Female,0,10,8,61
4,Non-Malignant,46,Male,0,8,22,50
5,Non-Malignant,73,Male,2,12,3,24
6,Malignant,60,Female,1,12,10,30
7,Non-Malignant,67,Male,2,12,15,35
8,Non-Malignant,43,Male,2,9,4,45
9,Non-Malignant,62,Male,0,5,6,37
10,Malignant,54,Male,2,4,60,63
11,Malignant,74,Male,1,9,37,51
12,Malignant,69,Male,1,11,10,NA
13,Malignant,70,Male,0,12,23,46
14,Malignant,64,Female,2,12,15,72
15,Non-Malignant,86,Female,0,10,3,29
16,Non-Malignant,42,Female,1,8,28,48
17,Malignant,26,Male,2,6,15,38
18,Non-Malignant,27,Female,1,8,NA,27
19,Non-Malignant,74,Female,0,11,8,21
20,Non-Malignant,49,Female,1,11,7,25
21,Malignant,52,Female,0,10,22,60
22,Non-Malignant,53,Male,1,10,11,10
23,Malignant,49,Male,2,4,14,30
24,Non-Malignant,30,Male,1,NA,17,34
25,Non-Malignant,57,Female,1,12,28,38
26,Non-Malignant,48,Female,2,3,17,31
27,Non-Malignant,60,Male,2,12,3,25
28,Non-Malignant,49,Female,0,9,15,48
29,Non-Malignant,64,Male,1,9,5,42
30,Malignant,54,Male,1,7,27,38
31,Non-Malignant,50,Male,1,11,21,57
32,Non-Malignant,81,Female,1,12,9,40
33,Non-Malignant,57,Female,1,9,10,32
34,Non-Malignant,72,Male,2,12,6,22
35,Non-Malignant,67,Male,1,9,NA,30
36,Non-Malignant,32,Male,1,3,13,14
37,Non-Malignant,81,Male,1,12,7,10
38,Non-Malignant,83,Female,0,8,13,68
39,Malignant,50,Female,1,12,9,26
40,Malignant,62,Male,2,9,16,65
41,Malignant,68,Female,0,11,25,64
42,Non-Malignant,68,Female,2,12,11,32
```

Line 46, Column 1 Tab Size: 4 Plain Text

SPACE DELIMITED WITHOUT HEADER

Cancer_Header.dat

```
Participant Diagnosis Age Gender Comorbids Optimism Depression VisImpair
1 Non-Malignant 43 Female 1 7 42 52
2 Non-Malignant 25 Female 1 6 3 43
3 Non-Malignant 55 Female 0 10 8 61
4 Non-Malignant 46 Male 0 8 22 50
5 Non-Malignant 73 Male 2 12 3 24
6 Malignant 60 Female 1 12 10 30
7 Non-Malignant 67 Male 2 12 15 35
8 Non-Malignant 43 Male 2 9 4 45
9 Non-Malignant 62 Male 0 5 6 37
10 Malignant 54 Male 2 4 60 63
11 Malignant 74 Male 1 9 37 51
12 Malignant 69 Male 1 11 10 NA
13 Malignant 70 Male 0 12 23 46
14 Malignant 64 Female 2 12 15 72
15 Non-Malignant 86 Female 0 10 3 29
16 Non-Malignant 42 Female 1 8 28 48
17 Malignant 26 Male 2 6 15 38
18 Non-Malignant 27 Female 1 8 NA 27
19 Non-Malignant 74 Female 0 11 8 21
20 Non-Malignant 49 Female 1 11 7 25
21 Malignant 52 Female 0 10 22 60
22 Non-Malignant 53 Male 1 10 11 10
23 Malignant 49 Male 2 4 14 30
24 Non-Malignant 30 Male 1 NA 17 34
25 Non-Malignant 57 Female 1 12 28 38
26 Non-Malignant 48 Female 2 3 17 31
27 Non-Malignant 60 Male 2 12 3 25
28 Non-Malignant 49 Female 0 9 15 48
29 Non-Malignant 64 Male 1 9 5 42
30 Malignant 54 Male 1 7 27 38
31 Non-Malignant 50 Male 1 11 21 57
32 Non-Malignant 81 Female 1 12 9 40
33 Non-Malignant 57 Female 1 9 10 32
34 Non-Malignant 72 Male 2 12 6 22
35 Non-Malignant 67 Male 1 9 NA 30
36 Non-Malignant 32 Male 1 3 13 14
37 Non-Malignant 81 Male 1 12 7 10
38 Non-Malignant 83 Female 0 8 13 68
39 Malignant 50 Female 1 12 9 26
40 Malignant 62 Male 2 9 16 65
41 Malignant 68 Female 0 11 25 64
```

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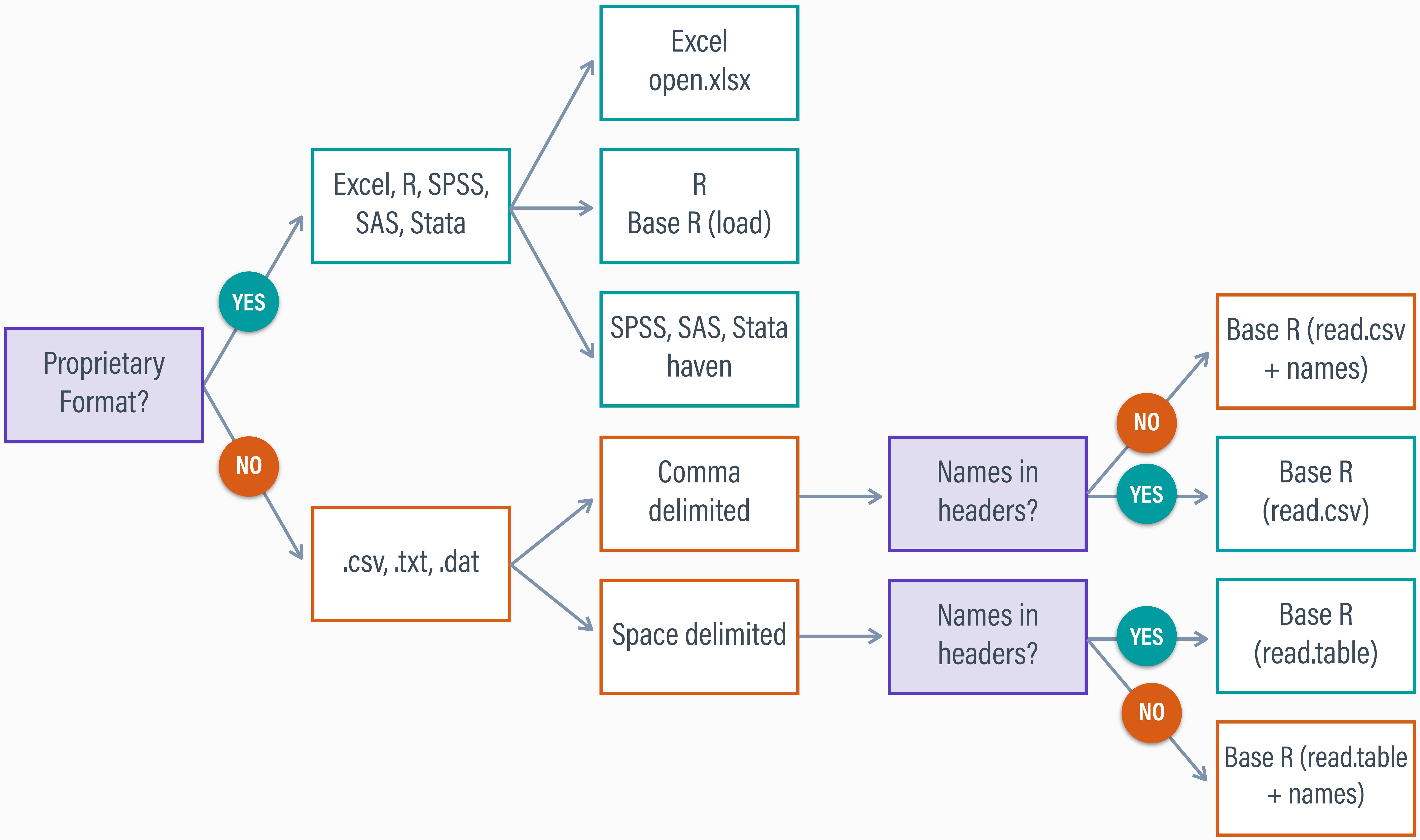
DATA IMPORT DECISION TREE



DEFINING A WORKING DIRECTORY

- Importing data requires a file path that identifies the location of the data on your local machine (usually long and complex)
- If you store your R script and data in the same directory, you can use the fdir package to get the file path for you
- fdir finds the folder path that contains your R script, and it sets R's default working directory to that location

SETTING THE WORKING DIRECTORY

```r
# LOAD R PACKAGES ----

library(fdir)       # sets the working directory to the location of this script
library(openxlsx)   # imports excel .xlsx files
library(haven)      # imports spss, sas, and stata data files

# SET THE DEFAULT WORKING DIRECTORY (FDIR) ----

# set working directory
set()

# print working directory
getwd()
```

IMPORTING EXCEL DATA

```
#----------------------------------------------------------------------#
# IMPORT PROPRIETARY FORMAT FROM EXCEL (OPENXLSX) ----
#----------------------------------------------------------------------#

# import excel .xlsx file from the working directory
mydata_excel <- read.xlsx('Cancer_Excel.xlsx', colNames = TRUE)

# convert string/character variables to R factors
mydata_excel[] <- lapply(mydata_excel, function(x) if (is.character(x)) factor(x) else x)

# summarize data
summary(mydata_excel)
```

IMPORTING R DATA

```
#----------------------------------------------------------------------#
# IMPORT PROPRIETARY FORMAT FROM R (BASE R) ----
#----------------------------------------------------------------------#

# import R .Rdata (or .rda) file from the working directory
load('Cancer_R.Rdata')

# summarize data
summary(mydata_R)
```

IMPORTING SPSS DATA

```
#-------------------------------------------------------------------#
# IMPORT PROPRIETARY FORMAT FROM SPSS (HAVEN) ----
#-------------------------------------------------------------------#

# import spss .sav file from the working directory
mydata_spss <- read_sav('Cancer_SPSS.sav')

# convert string/character variables to R factors
mydata_spss[] <- lapply(mydata_spss, function(x) if (is.character(x)) factor(x) else x)

# summarize data
summary(mydata_spss)
```

IMPORTING COMMA DELIMITED DATA

```
#------------------------------------------------------------------#
# IMPORT COMMA DELIMITED TEXT WITH NAMES IN HEADER (BASE R) ----
#------------------------------------------------------------------#

# import comma delimited .csv data with variable names from the working directory
mydata_csv1 <- read.csv('Cancer_Header.csv',     # file in working directory
                        stringsAsFactors = TRUE,    # convert text variables to R factors
                        header = TRUE,              # variable names in the first row
                        na.strings = c('NA', '999', '999.00'))     # missing data codes

# summarize data
summary(mydata_csv1)
```

IMPORTING COMMA DELIMITED DATA

```
#-------------------------------------------------------------------#
# IMPORT COMMA DELIMITED TEXT WITHOUT NAMES IN HEADER (BASE R) ----
#-------------------------------------------------------------------#

# import comma delimited .csv data without variable names from the working directory
mydata_csv2 <- read.csv('Cancer_NoHeader.csv',  # file in working directory
                    stringsAsFactors = TRUE,    # convert text variables to R factors
                    header = FALSE,             # no variable names in the first row
                    na.strings = c('NA', '999', '999.00'))  # missing data codes

# name data columns
names(mydata_csv2) <- c('Participant','Diagnosis','Age','Gender','Comorbids',
                    'Optimism','Depression','VisImpair')

# summarize data
summary(mydata_csv2)
```

IMPORTING SPACE DELIMITED DATA

```
#-------------------------------------------------------------------#
# IMPORT SPACE DELIMITED TEXT WITH NAMES IN HEADER (BASE R) ----
#-------------------------------------------------------------------#

# import space delimited .dat data with variable names from the working directory
mydata_dat1 <- read.table('Cancer_Header.dat',      # file in working directory
                         stringsAsFactors = TRUE,   # convert text variables to R factors
                         header = TRUE,             # variable names in the first row
                         sep = '',                  # whitespace separates variables
                         na.strings = c('NA', '999', '999.00'))  # missing data codes

# summarize data
summary(mydata_dat1)
```

IMPORTING SPACE DELIMITED DATA

```
#-----------------------------------------------------------------------#
# IMPORT SPACE DELIMITED TEXT WITHOUT NAMES IN HEADER (BASE R) ----
#-----------------------------------------------------------------------#

# import space delimited .dat data without variable names from the working directory
mydata_dat2 <- read.table('Cancer_NoHeader.dat',      # file in working directory
                          stringsAsFactors = TRUE,    # convert text variables to R factors
                          header = FALSE,             # no variable names in the first row
                          sep = '',                   # whitespace separates variables
                          na.strings = c('NA', '999', '999.00'))  # missing data codes

# name data columns
names(mydata_dat2) <- c('Participant','Diagnosis','Age','Gender','Comorbids',
                        'Optimism','Depression','VisImpair')

# summarize data
summary(mydata_dat2)
```